# Reduced vs Full model, Extra Sum of Squares, General Linear Hypothesis, Partial F test

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# Where We Left Off

# Tests So Far

- Overall significance: $H_0 : \beta_1 = \cdots = \beta_k = 0$ with $F_0 = \mathrm{MS_R}/\mathrm{MS_{Res}}$.
- Individual coefficient: $H_0 : \beta_j = 0$ with $t_0 = \hat{\beta}_j / \operatorname{se}(\hat{\beta}_j)$.
- Group of terms (Lecture 8): partial $F$-test comparing a full and a reduced model.
- Delivery time data, $n = 25$, $x_1$ = number of cases, $x_2$ = distance (ft): $\hat{y} = 2.341 + 1.616\, x_1 + 0.0144\, x_2$, $R^2 = 0.9596$.
- Overall test: $F_0 = 261.24$ on $(2, 22)$ degrees of freedom. Distance: $t_0 = 3.98$, $p = 0.0006$.

# Beyond One Coefficient at a Time

Questions that a single $t$-test or the overall $F$-test cannot answer:

- Does a *group* of regressors improve the fit once the others are already in the model?
- Do two regressors have the same effect, $\beta_j = \beta_k$?
- Is a coefficient, or a combination of coefficients, equal to a specified value?

All three are answered by comparing a *reduced* model with a *full* model. Lecture 8 used this comparison to test dummy variables and interactions; this lecture explains why the test works and extends it to all three questions.

# Reduced vs. Full Model

# Partitioning the Coefficient Vector

Model with $p$ parameters ($k = p - 1$ regressors):

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Split $\boldsymbol{\beta}$ and the columns of $\mathbf{X}$ into two blocks:

$$\boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_{(1)} \\ \boldsymbol{\beta}_{(2)} \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} \mathbf{X}_{(1)} & \mathbf{X}_{(2)} \end{bmatrix}$$

- $\boldsymbol{\beta}_{(1)}$ is $(p - q) \times 1$ and contains the intercept.
- $\boldsymbol{\beta}_{(2)}$ is $q \times 1$: the $q$ coefficients under test.

# The Full Model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_{(1)}\boldsymbol{\beta}_{(1)} + \mathbf{X}_{(2)}\boldsymbol{\beta}_{(2)} + \boldsymbol{\varepsilon}$$

- Uses all $p$ parameters.
- Least-squares fit: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.
- Residual sum of squares $\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})$ with $n - p$ degrees of freedom.
- Error variance estimate: $\mathrm{MS}_{\mathrm{Res}} = \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n - p)$.

# The Hypothesis and the Reduced Model

$$H_0 : \boldsymbol{\beta}_{(2)} = \mathbf{0} \qquad \text{versus} \qquad H_1 : \boldsymbol{\beta}_{(2)} \neq \mathbf{0}$$

If $H_0$ is true, the regressors in $\mathbf{X}_{(2)}$ drop out. The *reduced model* is

$$\mathbf{y} = \mathbf{X}_{(1)}\boldsymbol{\beta}_{(1)} + \boldsymbol{\varepsilon}$$

- Least-squares fit: $\hat{\boldsymbol{\beta}}_{(1)} = (\mathbf{X}'_{(1)}\mathbf{X}_{(1)})^{-1}\mathbf{X}'_{(1)}\mathbf{y}$.
- Residual sum of squares $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM})$ with $n - p + q$ degrees of freedom.

# Nested Models

- The reduced model is the full model with $\boldsymbol{\beta}_{(2)}$ forced to $\mathbf{0}$.
- Every fit the reduced model can produce is also available to the full model.
- The full model can only match or improve on the reduced model's fit.

$$\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) \;\geq\; \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})$$

The question is whether the improvement is larger than chance alone would produce.

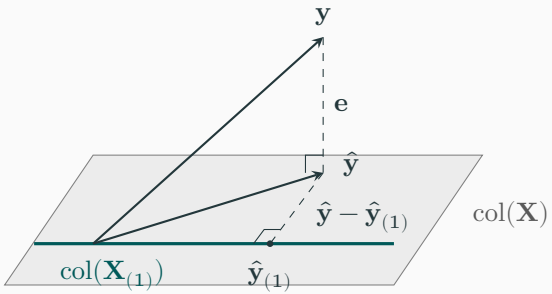
# Question: Why Not Just Compare the Residual Sums of Squares?

The full model always has the smaller (or equal) residual sum of squares.

- So can we conclude that $\mathbf{X}_{(2)}$ matters whenever $\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) < \mathrm{SS}_{\mathrm{Res}}(\mathrm{RM})$?
- What would a useful comparison need?

# Geometry of Nested Models

- $\mathrm{col}(\mathbf{X}_{(1)}) \subset \mathrm{col}(\mathbf{X})$.
- $\hat{\mathbf{y}}_{(1)} = \mathbf{H}_{(1)}\mathbf{y}$ projects onto the smaller space, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ onto the larger. Here $\mathbf{H}_{(1)} = \mathbf{X}_{(1)}(\mathbf{X}_{(1)}'\mathbf{X}_{(1)})^{-1}\mathbf{X}_{(1)}'$ is the hat matrix of the reduced model.
- $\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(1)}$ is the part of the fit due to $\mathbf{X}_{(2)}$.
- $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) = \|\mathbf{y} - \hat{\mathbf{y}}_{(1)}\|^2$ and $\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) = \|\mathbf{y} - \hat{\mathbf{y}}\|^2$.



# Delivery Time Data: Full vs. Reduced Model

Test whether distance ($x_2$) adds anything once cases ($x_1$) are in the model.

| Model | Fitted equation | $\mathrm{SS}_{\mathrm{Res}}$ | df |
|---|---|---|---|
| Full | $\hat{y} = 2.341 + 1.616\,x_1 + 0.0144\,x_2$ | 233.73 | 22 |
| Reduced | $\hat{y} = 3.321 + 2.176\,x_1$ | 402.13 | 23 |

- Here $\boldsymbol{\beta}_{(1)} = (\beta_0, \beta_1)'$ and $\boldsymbol{\beta}_{(2)} = \beta_2$, so $q = 1$.
- $\mathrm{MS}_{\mathrm{Res}}(\mathrm{FM}) = 233.73/22 = 10.62.$

# Extra Sum of Squares

# Extra Sum of Squares: Definition

Regression sum of squares of the full model, $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta})$ with $p-1$ df, and of the reduced model, $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(1)})$ with $p-q-1$ df. The *extra sum of squares* due to $\boldsymbol{\beta}_{(2)}$ given $\boldsymbol{\beta}_{(1)}$ is

$$\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)}) = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}) - \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(1)}) \tag{1}$$

with $q$ degrees of freedom.

# Interpreting the Extra Sum of Squares

- It is the increase in the regression sum of squares when $\mathbf{X}_{(2)}$ is added to a model that already contains $\mathbf{X}_{(1)}$.
- The word “given” matters: the contribution of $\mathbf{X}_{(2)}$ depends on what is already in the model.
- Equivalently, it is how much of the unexplained variation $\mathbf{X}_{(2)}$ removes.

# Extra Sum of Squares as a Reduction in $\mathrm{SS}_{\mathrm{Res}}$

Both models decompose the same total sum of squares:

$$\mathrm{SS}_{\mathrm{T}} = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}) + \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})$$
$$\mathrm{SS}_{\mathrm{T}} = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(1)}) + \mathrm{SS}_{\mathrm{Res}}(\mathrm{RM})$$

Subtracting the second line from the first and using (1):

$$\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)}) = \mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) \tag{2}$$

# Partitioning the Regression Sum of Squares

Rearranging (1):

$$\mathrm{SS_R}(\boldsymbol{\beta}) = \mathrm{SS_R}(\boldsymbol{\beta}_{(1)}) + \mathrm{SS_R}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)})$$

- The regression sum of squares splits into what $\mathbf{X}_{(1)}$ explains and what $\mathbf{X}_{(2)}$ adds.
- Degrees of freedom split the same way: $p - 1 = (p - q - 1) + q$.

# Delivery Time Data: Extra Sum of Squares for Distance

$$\begin{aligned}\mathrm{SS_R}(\beta_2 \mid \beta_1, \beta_0) &= \mathrm{SS_R}(\boldsymbol{\beta}) - \mathrm{SS_R}(\boldsymbol{\beta}_{(1)}) = 5550.81 - 5382.41 = 168.40 \\ &= \mathrm{SS_{Res}}(\mathrm{RM}) - \mathrm{SS_{Res}}(\mathrm{FM}) = 402.13 - 233.73 = 168.40\end{aligned}$$

- Both routes agree, as (2) requires.
- Adding distance to a model with cases reduces the residual sum of squares by 168.40, from 402.13 to 233.73.

# Distribution Under $H_0$

Assume $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$. If $H_0 : \boldsymbol{\beta}_{(2)} = \mathbf{0}$ is true:

- $\mathrm{SS_R}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)})/\sigma^2 \sim \chi^2_q$.
- $\mathrm{SS_{Res}}(\mathrm{FM})/\sigma^2 \sim \chi^2_{n-p}$.
- The two are independent.

The next frames show why, using the hat matrices of the two models.

# Projection Form of the Extra Sum of Squares

Hat matrices of the full and reduced models:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \qquad \mathbf{H}_{(1)} = \mathbf{X}_{(1)}(\mathbf{X}'_{(1)}\mathbf{X}_{(1)})^{-1}\mathbf{X}'_{(1)}$$

Residual sums of squares are quadratic forms in $\mathbf{y}$:

$$\begin{aligned} \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) &= \mathbf{y}'(\mathbf{I} - \mathbf{H})\mathbf{y} \\ \mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) &= \mathbf{y}'(\mathbf{I} - \mathbf{H}_{(1)})\mathbf{y} \\ \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)}) &= \mathbf{y}'(\mathbf{H} - \mathbf{H}_{(1)})\mathbf{y} \end{aligned}$$

# Why $\mathbf{H} - \mathbf{H}_{(1)}$ Is a Projection: Step 1

$\mathrm{col}(\mathbf{X}_{(1)}) \subset \mathrm{col}(\mathbf{X})$, so projecting a vector already in $\mathrm{col}(\mathbf{X}_{(1)})$ onto $\mathrm{col}(\mathbf{X})$ leaves it unchanged: $\mathbf{H}\mathbf{X}_{(1)} = \mathbf{X}_{(1)}$. Hence

$$\mathbf{H}\mathbf{H}_{(1)} = \mathbf{H}\mathbf{X}_{(1)}(\mathbf{X}'_{(1)}\mathbf{X}_{(1)})^{-1}\mathbf{X}'_{(1)} = \mathbf{X}_{(1)}(\mathbf{X}'_{(1)}\mathbf{X}_{(1)})^{-1}\mathbf{X}'_{(1)} = \mathbf{H}_{(1)}$$
$$\mathbf{H}_{(1)}\mathbf{H} = (\mathbf{H}\mathbf{H}_{(1)})' = \mathbf{H}_{(1)}$$

# Why $\mathbf{H} - \mathbf{H}_{(1)}$ Is a Projection: Step 2

$$(\mathbf{H} - \mathbf{H}_{(1)})^2 = \mathbf{H} - \mathbf{H}\mathbf{H}_{(1)} - \mathbf{H}_{(1)}\mathbf{H} + \mathbf{H}_{(1)} = \mathbf{H} - \mathbf{H}_{(1)}$$
$$\operatorname{rank}(\mathbf{H} - \mathbf{H}_{(1)}) = \operatorname{tr}(\mathbf{H} - \mathbf{H}_{(1)}) = p - (p - q) = q$$

- $\mathbf{H} - \mathbf{H}_{(1)}$ is symmetric and idempotent of rank $q$: a projection onto a $q$-dimensional space.
- Also $(\mathbf{H} - \mathbf{H}_{(1)})(\mathbf{I} - \mathbf{H}) = \mathbf{0}$, so the two quadratic forms involve orthogonal pieces of $\mathbf{y}$ and are independent.
- Under $H_0$, $\mathrm{E}[\mathbf{y}] = \mathbf{X}_{(1)}\boldsymbol{\beta}_{(1)}$ is annihilated (sent to $\mathbf{0}$) by $\mathbf{H} - \mathbf{H}_{(1)}$, so the form depends on $\boldsymbol{\varepsilon}$ only and is $\sigma^2\chi^2_q$.

# Background: What Does “Annihilate” Mean?

A matrix $\mathbf{A}$ *annihilates* a vector $\mathbf{v}$ if

$$\mathbf{A}\mathbf{v} = \mathbf{0}$$

- Nothing of $\mathbf{v}$ survives the multiplication.
- The set of all vectors that $\mathbf{A}$ annihilates is its *null space*.
- Familiar case: $\mathbf{I} - \mathbf{H}$ annihilates every vector in $\mathrm{col}(\mathbf{X})$, since $(\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta} - \mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. This is why the residuals $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$ contain no trace of $\mathrm{E}[\mathbf{y}]$.
- Here: $\mathbf{H}\mathbf{X}_{(1)} = \mathbf{X}_{(1)} = \mathbf{H}_{(1)}\mathbf{X}_{(1)}$, so $\mathbf{H} - \mathbf{H}_{(1)}$ annihilates $\mathrm{col}(\mathbf{X}_{(1)})$, which contains $\mathrm{E}[\mathbf{y}] = \mathbf{X}_{(1)}\boldsymbol{\beta}_{(1)}$ under $H_0$.

# Question: Why Does the Extra Sum of Squares Have $q$ Degrees of Freedom?

- We added $q$ columns to the model. How does that show up in the projection picture?

# Partial F-Test

# The Partial $F$ Statistic

$$F_0 = \frac{\mathrm{SS_R}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)})/q}{\mathrm{MS_{Res}}} \tag{3}$$

- $\mathrm{MS_{Res}}$ comes from the *full* model. It estimates $\sigma^2$ without bias whether or not $H_0$ is true; the reduced model's $\mathrm{MS_{Res}}$ overestimates $\sigma^2$ when $H_0$ is false, which would shrink $F_0$ exactly when we want to reject.
- Under $H_0 : \boldsymbol{\beta}_{(2)} = \mathbf{0}$, $F_0 \sim F_{q,\, n-p}$.
- Reject $H_0$ at level $\alpha$ if $F_0 > F_{\alpha,\, q,\, n-p}$.

# Partial $F$ Statistic from the Residual Sums of Squares

Substituting (2) into (3):

$$F_0 = \frac{[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})]/q}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n-p)} \tag{4}$$

Only the two residual sums of squares and their degrees of freedom are needed.

# Reading the Statistic

- Numerator: the average reduction in $\mathrm{SS}_{\mathrm{Res}}$ per added parameter.
- Denominator: the noise level $\sigma^2$, estimated from the full model.
- $F_0 \approx 1$: the added regressors reduce $\mathrm{SS}_{\mathrm{Res}}$ no more than noise would.
- $F_0$ large: the reduction is too big to be chance, so reject $H_0$.

# ANOVA Table with Extra-Sum-of-Squares Rows

| Source | SS | df | MS | $F_0$ |
|---|---|---|---|---|
| Regression | $\mathrm{SS_R}(\boldsymbol{\beta})$ | $p-1$ | $\mathrm{MS_R}$ | $\mathrm{MS_R}/\mathrm{MS_{Res}}$ |
| $\boldsymbol{\beta}_{(1)}$ | $\mathrm{SS_R}(\boldsymbol{\beta}_{(1)})$ | $p-q-1$ | | |
| $\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)}$ | $\mathrm{SS_R}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)})$ | $q$ | $\mathrm{SS_R}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)})/q$ | $\dfrac{\mathrm{SS_R}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)})/q}{\mathrm{MS_{Res}}}$ |
| Residual | $\mathrm{SS_{Res}}$ | $n-p$ | $\mathrm{MS_{Res}}$ | |
| Total | $\mathrm{SS_T}$ | $n-1$ | | |

# Special Case: The Overall $F$-Test

Take $\boldsymbol{\beta}_{(1)} = \beta_0$ and $\boldsymbol{\beta}_{(2)} = (\beta_1, \ldots, \beta_k)'$, so $q = p - 1 = k$.

- Reduced model: intercept only, $\hat{y}_i = \bar{y}$, so $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) = \mathrm{SS}_{\mathrm{T}}$.
- Extra sum of squares: $\mathrm{SS}_{\mathrm{T}} - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta})$.

$$F_0 = \frac{\mathrm{SS}_{\mathrm{R}}/k}{\mathrm{MS}_{\mathrm{Res}}} = \frac{\mathrm{MS}_{\mathrm{R}}}{\mathrm{MS}_{\mathrm{Res}}}$$

The overall test is the partial $F$-test with every slope in the tested block.

# Special Case: One Coefficient

Take $q = 1$, so $\boldsymbol{\beta}_{(2)} = \beta_j$. With $C_{jj} = [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$:

$$\mathrm{SS}_{\mathrm{R}}(\beta_j \mid \text{all other } \beta\text{'s}) = \frac{\hat{\beta}_j^2}{C_{jj}}$$

$$F_0 = \frac{\hat{\beta}_j^2 / C_{jj}}{\mathrm{MS}_{\mathrm{Res}}} = \left( \frac{\hat{\beta}_j}{\sqrt{\mathrm{MS}_{\mathrm{Res}}\, C_{jj}}} \right)^2 = t_0^2$$

The partial $F$-test for a single coefficient is the two-sided $t$-test, since $F_{1,\nu} = t_\nu^2$.

Why $\hat{\beta}_j^2 / C_{jj}$: this is the matrix form (6) of the general linear hypothesis (later in this lecture) with $\mathbf{T}$ the row that has $1$ in the position of $\beta_j$ and $0$ elsewhere, so $\mathbf{T}\hat{\boldsymbol{\beta}} = \hat{\beta}_j$ and $\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}' = C_{jj}$.

# Delivery Time Data: Does Distance Add to Cases?

$H_0 : \beta_2 = 0$ versus $H_1 : \beta_2 \neq 0$, given $x_1$ in the model.

| | $\mathrm{SS}_{\mathrm{Res}}$ | df |
|---|---|---|
| Full: $x_1, x_2$ | 233.73 | 22 |
| Reduced: $x_1$ | 402.13 | 23 |
| Difference | 168.40 | 1 |

# Delivery Time Data: Partial $F$ for Distance

$$F_0 = \frac{168.40/1}{233.73/22} = \frac{168.40}{10.62} = 15.85$$

- $F_{0.05,1,22} = 4.30$ and $p = 0.0006$: reject $H_0$.
- Distance contributes significantly given cases.
- Check: $t_0 = 3.98$ for $\beta_2$ and $t_0^2 = 15.85 = F_0$.

# Delivery Time Data: Does Cases Add to Distance?

Reverse the roles: $H_0 : \beta_1 = 0$, with $x_2$ in the reduced model.

$$F_0 = \frac{(1185.39 - 233.73)/1}{233.73/22} = \frac{951.66}{10.62} = 89.58$$

- $p = 3.3 \times 10^{-9}$: reject $H_0$.
- Check: $t_0 = 9.46$ for $\beta_1$ and $t_0^2 = 89.58$.

# A Second Dataset: Gasoline Mileage

MPV Table B.3: 32 automobiles, response $y$ = miles per gallon.

- Regressors used here: horsepower, torque, rear axle ratio, carburetor barrels, weight.
- Torque is missing for two cars, so we use the $n = 30$ complete cases.

$$y = \beta_0 + \beta_{\mathrm{hp}} x_{\mathrm{hp}} + \beta_{\mathrm{tq}} x_{\mathrm{tq}} + \beta_{\mathrm{ax}} x_{\mathrm{ax}} + \beta_{\mathrm{cb}} x_{\mathrm{cb}} + \beta_{\mathrm{wt}} x_{\mathrm{wt}} + \varepsilon$$

Here $p = 6$, so the residual degrees of freedom are $n - p = 24$.

# Gasoline Mileage: Full Model Fit

| Term | Estimate | Std. error | $t_0$ | $p$-value |
|---|---|---|---|---|
| Intercept | 29.2464 | 7.2112 | 4.056 | 0.0005 |
| Horsepower | $-0.0115$ | 0.0767 | $-0.149$ | 0.882 |
| Torque | $-0.0223$ | 0.0526 | $-0.423$ | 0.676 |
| Rear axle ratio | 2.1197 | 2.1420 | 0.990 | 0.332 |
| Carburetor barrels | 0.0425 | 1.1541 | 0.037 | 0.971 |
| Weight | $-0.00261$ | 0.00200 | $-1.303$ | 0.205 |

$\mathrm{SS}_{\mathrm{R}} = 858.19$ (5 df), $\mathrm{SS}_{\mathrm{Res}} = 280.91$ (24 df), $\mathrm{MS}_{\mathrm{Res}} = 11.705$, $R^2 = 0.753$, overall $F_0 = 14.66$ ($p \approx 10^{-6}$).

# Question: Every $t$-Test Is Insignificant, Yet the Overall $F$ Is Large

- How can the model be highly significant while no individual coefficient is?
- Sample correlations: horsepower–torque $0.96$, torque–weight $0.91$, horsepower–weight $0.85$.

# Gasoline Mileage: Testing a Group of Regressors

$H_0 : \beta_{\text{hp}} = \beta_{\text{tq}} = \beta_{\text{wt}} = 0$, so $q = 3$: do the three engine-size and weight measures add anything beyond axle ratio and carburetor?

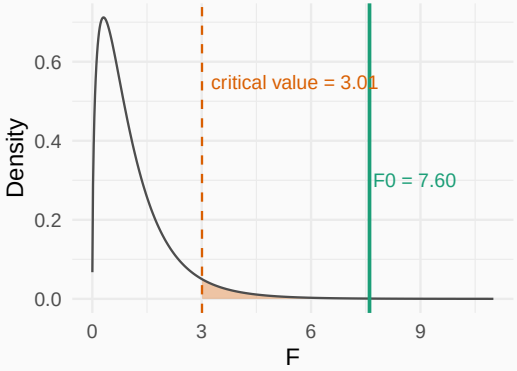
| | $\text{SS}_{\text{Res}}$ | df |
|---|---|---|
| Full: hp, tq, ax, cb, wt | 280.91 | 24 |
| Reduced: ax, cb | 547.87 | 27 |
| Difference | 266.96 | 3 |

Reduced fit: $\widehat{y} = 5.054 + 6.611\, x_{\text{ax}} - 2.041\, x_{\text{cb}}$.

# Gasoline Mileage: Partial $F$ Calculation

$$F_0 = \frac{266.96/3}{280.91/24} = \frac{88.99}{11.705} = 7.60$$

- $F_{0.05,3,24} = 3.01$.
- $p = 0.001$: reject $H_0$.
- The group matters even though each member has $p > 0.2$ individually.



# Gasoline Mileage: A Group That Does Not Help

$H_0 : \beta_{\text{ax}} = \beta_{\text{cb}} = 0$, so $q = 2$, with hp, tq, wt kept in the model.

$$F_0 = \frac{12.74/2}{280.91/24} = \frac{6.368}{11.705} = 0.54$$

- $F_{0.05,2,24} = 3.40$ and $p = 0.59$: do not reject $H_0$.
- Given horsepower, torque and weight, axle ratio and carburetor add nothing detectable.

# Question: Why One Partial $F$ Instead of Several $t$-Tests?

- For $H_0 : \beta_{\text{hp}} = \beta_{\text{tq}} = \beta_{\text{wt}} = 0$ we could run three $t$-tests. Why is that a poor substitute for $F_0 = 7.60$?

# General Linear Hypothesis

# Linear Restrictions on the Coefficients

$$H_0 : \mathbf{T}\boldsymbol{\beta} = \mathbf{0} \qquad \text{versus} \qquad H_1 : \mathbf{T}\boldsymbol{\beta} \neq \mathbf{0}$$

- $\mathbf{T}$ is a $q \times p$ matrix of rank $q$.
- Each row of $\mathbf{T}$ is one linear restriction on $\boldsymbol{\beta}$.
- Rank $q$ means the $q$ restrictions are independent: none is implied by the others.

# Examples of $\mathbf{T}$ for the Gasoline Model

Parameter order $(\beta_0, \beta_{\text{hp}}, \beta_{\text{tq}}, \beta_{\text{ax}}, \beta_{\text{cb}}, \beta_{\text{wt}})'$.

| Hypothesis | $q$ | $\mathbf{T}$ |
|---|---|---|
| $\beta_{\text{wt}} = 0$ | 1 | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ |
| $\beta_{\text{hp}} = \beta_{\text{tq}} = 0$ | 2 | $\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$ |
| $\beta_{\text{hp}} = \beta_{\text{tq}}$ | 1 | $\begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 \end{bmatrix}$ |
| $\beta_{\text{hp}} = \beta_{\text{tq}},\ \beta_{\text{ax}} = 0$ | 2 | $\begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$ |

# Subset Tests Are a Special Case

If the tested block $\boldsymbol{\beta}_{(2)}$ is placed last:

$$\mathbf{T} = \begin{bmatrix} \mathbf{0}_{q\times(p-q)} & \mathbf{I}_q \end{bmatrix} \quad \Longrightarrow \quad \mathbf{T}\boldsymbol{\beta} = \boldsymbol{\beta}_{(2)}$$

- $H_0 : \mathbf{T}\boldsymbol{\beta} = \mathbf{0}$ is exactly $H_0 : \boldsymbol{\beta}_{(2)} = \mathbf{0}$.
- Single-coefficient, subset and overall tests are all general linear hypotheses.
- The general form also covers restrictions that are not of the form “coefficients equal zero”.

# Obtaining the Reduced Model

Impose the restrictions in the full model and simplify.

Example: $H_0 : \beta_{\text{hp}} = \beta_{\text{tq}} = \gamma$. Substituting,

$$y = \beta_0 + \gamma\,(x_{\text{hp}} + x_{\text{tq}}) + \beta_{\text{ax}} x_{\text{ax}} + \beta_{\text{cb}} x_{\text{cb}} + \beta_{\text{wt}} x_{\text{wt}} + \varepsilon$$

- The two columns are replaced by their sum.
- The reduced model has $p - q = 5$ parameters and is fit by ordinary least squares.

# Test Statistic for the General Linear Hypothesis

$$F_0 = \frac{[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})]/q}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n-p)} \tag{5}$$

- Same form as (4); $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM})$ is now the residual sum of squares under the restrictions.
- Under $H_0$, $F_0 \sim F_{q,\,n-p}$. Reject if $F_0 > F_{\alpha,\,q,\,n-p}$.

# Matrix Form of the Test

The difference in residual sums of squares can be computed from the full-model fit alone:

$$\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) = \mathrm{SS}_{\mathrm{H}} = (\mathbf{T}\hat{\boldsymbol{\beta}})' \left[\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\right]^{-1} (\mathbf{T}\hat{\boldsymbol{\beta}}) \qquad (6)$$

- $F_0 = (\mathrm{SS}_{\mathrm{H}}/q)/\mathrm{MS}_{\mathrm{Res}}$, the same statistic as (5).
- No reduced model needs to be fit.

The next two frames derive (6).

# Deriving $\mathrm{SS_H}$, Step 1: Restricted Least Squares

The reduced-model fit $\tilde{\boldsymbol{\beta}}$ minimizes $(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})$ subject to $\mathbf{T}\boldsymbol{\beta}=\mathbf{0}$. With a Lagrange multiplier vector $\lambda$ ($q\times 1$), set the derivative of

$$(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})+2\lambda'\mathbf{T}\boldsymbol{\beta}$$

to zero: $\mathbf{X}'\mathbf{X}\tilde{\boldsymbol{\beta}}=\mathbf{X}'\mathbf{y}-\mathbf{T}'\lambda$, so $\tilde{\boldsymbol{\beta}}=\hat{\boldsymbol{\beta}}-(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\lambda$. Imposing $\mathbf{T}\tilde{\boldsymbol{\beta}}=\mathbf{0}$ gives $\lambda=\left[\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\right]^{-1}\mathbf{T}\hat{\boldsymbol{\beta}}$, hence

$$\tilde{\boldsymbol{\beta}}=\hat{\boldsymbol{\beta}}-(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\left[\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\right]^{-1}\mathbf{T}\hat{\boldsymbol{\beta}} \tag{7}$$

# Deriving $\mathrm{SS}_{\mathrm{H}}$, Step 2: The Residual Sum of Squares

Write the reduced-model residuals as $\mathbf{y} - \mathbf{X}\tilde{\boldsymbol{\beta}} = \mathbf{e} + \mathbf{X}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}})$. The cross term vanishes because $\mathbf{X}'\mathbf{e} = \mathbf{0}$:

$$\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) = \mathbf{e}'\mathbf{e} + (\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}})'\mathbf{X}'\mathbf{X}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}})$$

Let $\mathbf{M} = \mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'$. By (7), $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\mathbf{M}^{-1}\mathbf{T}\hat{\boldsymbol{\beta}}$, so

$$\begin{aligned}\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) &= (\mathbf{T}\hat{\boldsymbol{\beta}})'\mathbf{M}^{-1}\,\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\,\mathbf{M}^{-1}(\mathbf{T}\hat{\boldsymbol{\beta}}) \\ &= (\mathbf{T}\hat{\boldsymbol{\beta}})'\mathbf{M}^{-1}\mathbf{M}\mathbf{M}^{-1}(\mathbf{T}\hat{\boldsymbol{\beta}}) = (\mathbf{T}\hat{\boldsymbol{\beta}})'\mathbf{M}^{-1}(\mathbf{T}\hat{\boldsymbol{\beta}}) = \mathrm{SS}_{\mathrm{H}}\end{aligned}$$

# Why the Quadratic Form Works

- $\mathrm{Var}(\mathbf{T}\hat{\boldsymbol{\beta}}) = \sigma^2\,\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'$.
- Under $H_0$, $\mathbf{T}\hat{\boldsymbol{\beta}}$ has mean $\mathbf{0}$, so $\mathrm{SS}_{\mathrm{H}}/\sigma^2$ is a standardized squared length of a $q$-dimensional normal vector: $\chi^2_q$.
- $\mathrm{SS}_{\mathrm{H}}$ depends on the data only through $\hat{\boldsymbol{\beta}}$, and $\hat{\boldsymbol{\beta}}$ is independent of $\mathrm{MS}_{\mathrm{Res}}$ (Lecture 5). So $F_0 = (\mathrm{SS}_{\mathrm{H}}/q)/\mathrm{MS}_{\mathrm{Res}} \sim F_{q,\,n-p}$ under $H_0$.
- For $q = 1$, $\mathrm{SS}_{\mathrm{H}}/\mathrm{MS}_{\mathrm{Res}}$ reduces to $(\mathbf{T}\hat{\boldsymbol{\beta}})^2/(\mathrm{MS}_{\mathrm{Res}}\,\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}') = t_0^2$.

# Gasoline Mileage: $H_0 : \beta_{\text{hp}} = \beta_{\text{tq}}$ by the Matrix Form

$\mathbf{T} = \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 \end{bmatrix}$, $q = 1$.

$$\begin{aligned} \mathbf{T}\hat{\boldsymbol{\beta}} &= \hat{\beta}_{\text{hp}} - \hat{\beta}_{\text{tq}} = -0.01146 - (-0.02228) = 0.01083 \\ \mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}' &= 0.001278 \\ \text{SS}_{\text{H}} &= \frac{0.01083^2}{0.001278} = 0.0917 \\ F_0 &= \frac{0.0917/1}{11.705} = 0.0078 \end{aligned}$$

$F_{0.05,1,24} = 4.26$ and $p = 0.930$: do not reject.

# Gasoline Mileage: $H_0 : \beta_{\text{hp}} = \beta_{\text{tq}}$ by the Reduced Model

Fit $y = \beta_0 + \gamma(x_{\text{hp}} + x_{\text{tq}}) + \beta_{\text{ax}} x_{\text{ax}} + \beta_{\text{cb}} x_{\text{cb}} + \beta_{\text{wt}} x_{\text{wt}}$:

$$\begin{aligned} \widehat{y} &= 29.061 - 0.01800\,(x_{\text{hp}} + x_{\text{tq}}) \\ &\quad + 2.219\,x_{\text{ax}} + 0.0699\,x_{\text{cb}} - 0.002673\,x_{\text{wt}} \\ \mathrm{SS}_{\text{Res}}(\text{RM}) &= 281.005 \quad (25\ \text{df}) \end{aligned}$$

- $\mathrm{SS}_{\text{Res}}(\text{RM}) - \mathrm{SS}_{\text{Res}}(\text{FM}) = 281.005 - 280.914 = 0.0917$, matching $\mathrm{SS}_{\text{H}}$.
- We cannot reject equality, but that is weak evidence *for* it. Horsepower and torque are correlated at $0.96$, so $\widehat{\beta}_{\text{hp}} - \widehat{\beta}_{\text{tq}} = 0.0108$ has standard error $\sqrt{11.705 \times 0.001278} = 0.122$ and 95% interval $(-0.24,\ 0.26)$, far wider than either coefficient. The test has little power.

# Gasoline Mileage: Two Restrictions at Once

$H_0 : \beta_{\text{hp}} = \beta_{\text{tq}}$ and $\beta_{\text{ax}} = 0$, so $q = 2$. The reduced model has three predictor terms:

$$y = \beta_0 + \gamma\,(x_{\text{hp}} + x_{\text{tq}}) + \beta_{\text{cb}} x_{\text{cb}} + \beta_{\text{wt}} x_{\text{wt}} + \varepsilon$$

$$\text{SS}_{\text{Res}}(\text{RM}) = 298.35 \ (26 \text{ df})$$

$$F_0 = \frac{(298.35 - 280.91)/2}{11.705} = \frac{8.718}{11.705} = 0.74$$

$F_{0.05,2,24} = 3.40$ and $p = 0.49$: do not reject.

# Nonzero Right-Hand Side: $\mathbf{T}\beta = \mathbf{c}$

$$H_0 : \mathbf{T}\boldsymbol{\beta} = \mathbf{c} \qquad \Longrightarrow \qquad \mathrm{SS}_{\mathrm{H}} = (\mathbf{T}\hat{\boldsymbol{\beta}} - \mathbf{c})' \left[\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\right]^{-1} (\mathbf{T}\hat{\boldsymbol{\beta}} - \mathbf{c})$$

The derivation is the same as before, with $\mathbf{T}\hat{\boldsymbol{\beta}}$ replaced by $\mathbf{T}\hat{\boldsymbol{\beta}} - \mathbf{c}$.
Reduced model: move the known part to the left-hand side.

- Example: $H_0 : \beta_{\mathrm{cb}} = 1$. Regress $y - x_{\mathrm{cb}}$ on the other regressors: $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) = 288.97$.
- $F_0 = (288.97 - 280.91)/11.705 = 0.69$ and $p = 0.41$.
- Check: $t_0 = (0.0425 - 1)/1.1541 = -0.830$ and $t_0^2 = 0.69$.

# Summary of Special Cases

| Hypothesis | $\mathbf{T}$ rows | $q$ | Reduced model |
|---|---|---|---|
| Single coefficient | one unit row | 1 | Drop the regressor |
| Subset | $q$ unit rows | $q$ | Drop the subset |
| All slopes | $k$ unit rows | $k$ | Intercept only |
| Equal coefficients | $(\dots, 1, \dots, -1, \dots)$ | 1 | Sum the two columns |
| Several restrictions | $q$ independent rows | $q$ | Substitute all restrictions |

All use (5) with $F_{q,\, n-p}$.

# Question: Is $\beta_{\text{hp}} = \beta_{\text{tq}}$ Meaningful?

- Horsepower and torque are on different scales. Is a test of equal coefficients meaningful?
- How could the hypothesis be made scale-free?

# Sequential vs. Partial Sums of Squares

# Sequential Sums of Squares

Enter the regressors in a chosen order $x_1, x_2, \dots, x_k$. The sequential sums of squares are

$$\mathrm{SS_R}(\beta_1 \mid \beta_0), \quad \mathrm{SS_R}(\beta_2 \mid \beta_0, \beta_1), \quad \dots, \quad \mathrm{SS_R}(\beta_k \mid \beta_0, \dots, \beta_{k-1})$$

- Each is an extra sum of squares with $q = 1$.
- Each regressor is adjusted only for those *entered before it*.
- Software often calls these Type I sums of squares.

# Sequential Sums of Squares Add Up

Applying (1) one regressor at a time telescopes:

$$\mathrm{SS_R}(\boldsymbol{\beta}) = \sum_{j=1}^{k} \mathrm{SS_R}(\beta_j \mid \beta_0, \ldots, \beta_{j-1}) \tag{8}$$

- The decomposition accounts for the whole regression sum of squares.
- It depends on the order of entry.

# Partial Sums of Squares

$$\mathrm{SS}_{\mathrm{R}}(\beta_j \mid \beta_0, \dots, \beta_{j-1}, \beta_{j+1}, \dots, \beta_k)$$

- Each regressor is adjusted for *all* the others, as if it entered last.
- Equal to $t_0^2 \, \mathrm{MS}_{\mathrm{Res}}$ for the individual $t$-test, hence the partial $F$-test with $q = 1$.
- Software often calls these Type III sums of squares.
- In general they do not add up to $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta})$.

# Delivery Time Data: Sequential vs. Partial

| | Sequential | | Partial |
|---|---|---|---|
| | $x_1$ first | $x_2$ first | |
| Cases ($x_1$) | 5382.41 | 951.66 | 951.66 |
| Distance ($x_2$) | 168.40 | 4599.15 | 168.40 |
| Sum | 5550.81 | 5550.81 | 1120.06 |

- Both sequential columns add to $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}) = 5550.81$; the partial column does not.
- The size of each regressor's share depends on the order.

# Question: Which Sequential Sum of Squares Equals the Partial One?

- In the delivery table, which sequential entries coincide with the partial sums of squares?
- Why?

# Gasoline Mileage: Sequential vs. Partial

| | Sequential | | Partial |
|---|---|---|---|
| | hp, tq, ax, cb, wt | wt, cb, ax, tq, hp | |
| Horsepower | 723.26 | 0.26 | 0.26 |
| Torque | 107.17 | 8.79 | 2.10 |
| Rear axle ratio | 2.78 | 59.15 | 11.46 |
| Carburetor | 5.11 | 10.68 | 0.02 |
| Weight | 19.87 | 779.31 | 19.87 |
| Sum | 858.19 | 858.19 | 33.71 |

Horsepower carries 723 when entered first and 0.26 when entered last: the regressors overlap heavily.

# When Sequential Equals Partial: Orthogonal Columns

If $\mathbf{X}_{(1)}'\mathbf{X}_{(2)} = \mathbf{0}$, then $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(2)} \mid \boldsymbol{\beta}_{(1)}) = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(2)})$: the contribution of each block is unaffected by the other, in any order. Here $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_{(2)})$ is the regression sum of squares when $y$ is regressed on $\mathbf{X}_{(2)}$ alone (plus an intercept). Since $\mathbf{X}_{(1)}$ contains the column of ones, $\mathbf{X}_{(1)}'\mathbf{X}_{(2)} = \mathbf{0}$ also requires every column of $\mathbf{X}_{(2)}$ to have mean zero.

- Delivery data: replace $x_2$ by $z = e(x_2 \mid x_1)$, the part of distance not explained by cases (orthogonal to $\mathbf{1}$ and $x_1$).
- Regress $y$ on $x_1$ and $z$: sequential sums of squares are $5382.41$ for $x_1$ and $168.40$ for $z$ in either order.
- $168.40$ is exactly $\mathrm{SS}_{\mathrm{R}}(\beta_2 \mid \beta_1, \beta_0)$ from the original fit.

The same $z$ will reappear as the horizontal axis of an added-variable plot.

# Software Output

| | Sequential | Partial |
|---|---|---|
| R | `anova(fit)` | `drop1(fit, test = "F")`<br>`car::Anova(fit, type = "III")` |

- `anova()` depends on the order of terms in the model formula.
- The $F$-statistics in the partial output equal $t_0^2$ for each coefficient.
- Check which type your software reports by default before interpreting a table.

# Reading the $F$-Tests in `anova(fit)`

Row $j$ of the sequential table reports

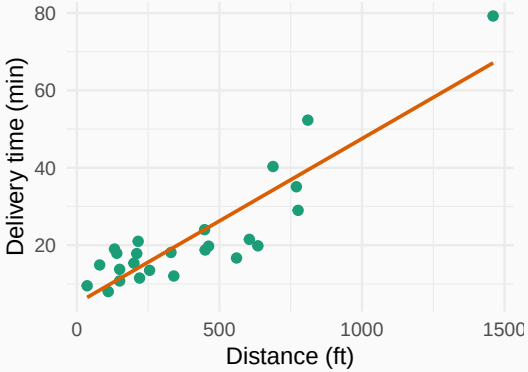
$$F_0 = \frac{\mathrm{SS_R}(\beta_j \mid \beta_0, \ldots, \beta_{j-1})/1}{\mathrm{MS_{Res}}}, \qquad \mathrm{MS_{Res}} \text{ from the full model.}$$

- It asks whether $x_j$ adds to the regressors entered *before* it, ignoring those entered after it.
- Only the last row is a partial $F$-test; the other rows change when the order changes.
- Delivery data, cases first: $F_0 = 5382.41/10.62 = 506.6$ for cases and $15.85$ for distance (the partial $F$).
- Distance first: $F_0 = 432.9$ for distance and $89.58$ for cases (the partial $F$).

# Added-Variable Plots

# Why Look at a Plot?

- A scatterplot of $y$ against $x_j$ ignores the other regressors.
- Delivery data: the marginal slope of time on distance is $0.0426$ min/ft.
- The coefficient of distance in the full model is $0.0144$ min/ft.
- We want a plot showing the effect of $x_j$ *after* adjusting for the rest.



# Constructing an Added-Variable Plot

Let $\mathbf{X}_{(-j)}$ be the design matrix with the column of $x_j$ removed (intercept kept).

1. Regress $y$ on $\mathbf{X}_{(-j)}$ and keep the residuals $e(y \mid \mathbf{X}_{(-j)})$.
2. Regress $x_j$ on $\mathbf{X}_{(-j)}$ and keep the residuals $e(x_j \mid \mathbf{X}_{(-j)})$.
3. Plot $e(y \mid \mathbf{X}_{(-j)})$ against $e(x_j \mid \mathbf{X}_{(-j)})$.

Also called a partial regression plot.

# Residuals as Adjusted Variables

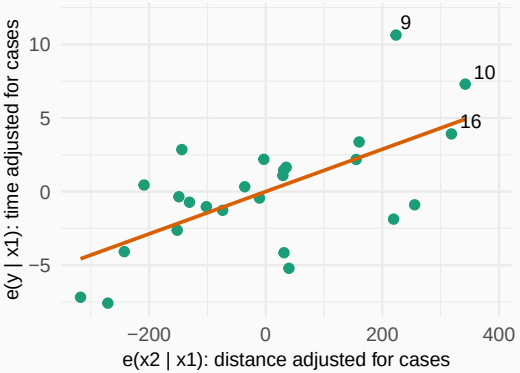
With $\mathbf{H}_{(-j)} = \mathbf{X}_{(-j)}(\mathbf{X}'_{(-j)}\mathbf{X}_{(-j)})^{-1}\mathbf{X}'_{(-j)}$:

$$e(y \mid \mathbf{X}_{(-j)}) = (\mathbf{I} - \mathbf{H}_{(-j)})\mathbf{y}$$
$$e(x_j \mid \mathbf{X}_{(-j)}) = (\mathbf{I} - \mathbf{H}_{(-j)})\mathbf{x}_j$$

- The first is the part of $y$ not explained by the other regressors.
- The second is the part of $x_j$ not explained by the other regressors.

# Delivery Time Data: Added-Variable Plot for Distance

- Adjusted for cases, longer distances go with longer delivery times.
- The fitted line has slope $0.01438$, the coefficient $\hat{\beta}_2$ from the full model.
- Labelled points: 9 has the largest residual; 10 and 16 have the most extreme adjusted distance.



# Key Property: The Slope Is $\widehat{\beta}_j$

The least-squares line through the added-variable plot has:

- Slope $\widehat{\beta}_j$, the coefficient from the full model.
- Intercept $0$.
- Residuals equal to the full-model residuals $\mathbf{e}$.

The next two frames show why.

# Derivation, Step 1: Decompose $y$

The full-model fit separates $x_j$ from the other regressors:

$$\mathbf{y} = \mathbf{X}_{(-j)}\hat{\boldsymbol{\beta}}_{(-j)} + \mathbf{x}_j\hat{\beta}_j + \mathbf{e}$$

Apply $\mathbf{I} - \mathbf{H}_{(-j)}$. It annihilates $\mathbf{X}_{(-j)}$, and $(\mathbf{I} - \mathbf{H}_{(-j)})\mathbf{e} = \mathbf{e}$ because $\mathbf{e} \perp \mathrm{col}(\mathbf{X}) \supset \mathrm{col}(\mathbf{X}_{(-j)})$:

$$e(y \mid \mathbf{X}_{(-j)}) = \hat{\beta}_j \, e(x_j \mid \mathbf{X}_{(-j)}) + \mathbf{e} \tag{9}$$

# Derivation, Step 2: Slope and Residuals

Both adjusted variables have mean zero, so the line through the plot has no intercept. Its slope is

$$\frac{e(x_j \mid \mathbf{X}_{(-j)})' \, e(y \mid \mathbf{X}_{(-j)})}{e(x_j \mid \mathbf{X}_{(-j)})' \, e(x_j \mid \mathbf{X}_{(-j)})} = \hat{\beta}_j + \frac{e(x_j \mid \mathbf{X}_{(-j)})' \, \mathbf{e}}{e(x_j \mid \mathbf{X}_{(-j)})' \, e(x_j \mid \mathbf{X}_{(-j)})}$$

The numerator of the last term is $\mathbf{x}_j'(\mathbf{I} - \mathbf{H}_{(-j)})\mathbf{e} = \mathbf{x}_j'\mathbf{e} = 0$. So the slope is $\hat{\beta}_j$ and, by (9), the residuals about the line are $\mathbf{e}$.

# Consequences

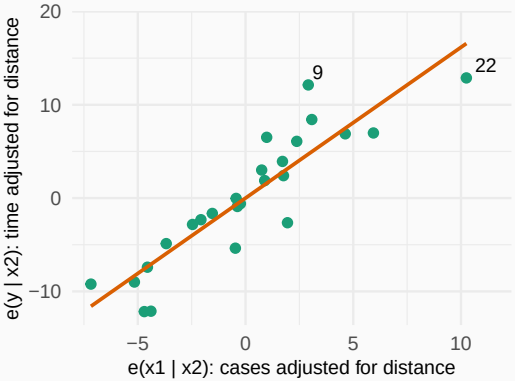
- The plot has the same residuals, hence the same residual sum of squares, as the full model.
- The sum of squares explained by the line is $\hat{\beta}_j^2 \sum_i e_i(x_j \mid \mathbf{X}_{(-j)})^2 = \mathrm{SS_R}(\beta_j \mid \text{others})$. Why: square both sides of (9); the cross term is $0$ because $e(x_j \mid \mathbf{X}_{(-j)})'\mathbf{e} = 0$, so

$$\underbrace{\sum e(y \mid \mathbf{X}_{(-j)})^2}_{\mathrm{SS_{Res}}(\text{without } x_j)} = \hat{\beta}_j^2 \sum e(x_j \mid \mathbf{X}_{(-j)})^2 + \underbrace{\mathbf{e}'\mathbf{e}}_{\mathrm{SS_{Res}}(\mathrm{FM})},$$

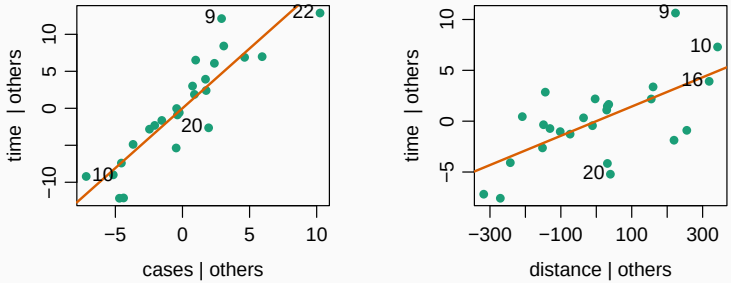
  and the difference of the two residual sums of squares is the extra sum of squares (2).
- Delivery data, distance: $0.014385^2 \times 813{,}838.7 = 168.40$.
- Outliers and influential points for $\hat{\beta}_j$ can be seen directly in the plot.

# Delivery Time Data: Added-Variable Plot for Cases

- Slope $1.616$, the full-model coefficient $\hat{\beta}_1$.
- The sum of squares due to the line is $951.66 = \mathrm{SS}_{\mathrm{R}}(\beta_1 \mid \beta_2, \beta_0)$.
- Point 22 has the most extreme adjusted number of cases.
- Point 9 again has the largest residual.



# Added-Variable Plots in R: `car::avPlots()`




- `car::avPlots(fit)` draws one panel per regressor, the same plots as the previous two slides.
- By default it labels the two largest residuals (9, 20) and the two most extreme points horizontally in each panel.

# Squared Correlation in the Plot

The squared correlation between the two adjusted variables is

$$r^2 = \frac{\hat{\beta}_j^2 \sum e(x_j \mid \mathbf{X}_{(-j)})^2}{\sum e(y \mid \mathbf{X}_{(-j)})^2} = \frac{\mathrm{SS}_{\mathrm{R}}(\beta_j \mid \text{others})}{\mathrm{SS}_{\mathrm{Res}}(\text{model without } x_j)}$$

- Distance: $168.40/402.13 = 0.419$, correlation $0.647$.
- Cases: $951.66/1185.39 = 0.803$, correlation $0.896$.
- This is the partial correlation of $y$ and $x_j$ given the other regressors; it returns in model selection.

# Collinearity in the Plot

The horizontal spread of the plot is what is left of $x_j$ after the other regressors take their share:

$$\sum_i e_i(x_j \mid \mathbf{X}_{(-j)})^2 = (1-R_j^2)\sum_i (x_{ij}-\bar{x}_j)^2, \qquad \mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\sum_i e_i(x_j \mid \mathbf{X}_{(-j)})^2}$$

- $R_j^2$ is the $R^2$ from regressing $x_j$ on the other regressors.
- Delivery data: $\sum_i (x_{i2} - \bar{x}_2)^2 = 2{,}537{,}935$ for distance falls to $\sum_i e_i(x_2 \mid x_1)^2 = 813{,}838.7$ after adjusting for cases, a fraction $0.321 = 1 - 0.824^2$, where $0.824 = \mathrm{Corr}(x_1, x_2)$.
- $1/(1 - R_j^2)$ is the variance inflation factor (VIF), covered next lecture; here $1/0.321 = 3.12$.
- Strong collinearity gives a narrow plot and an imprecise slope.

# What to Look for in an Added-Variable Plot

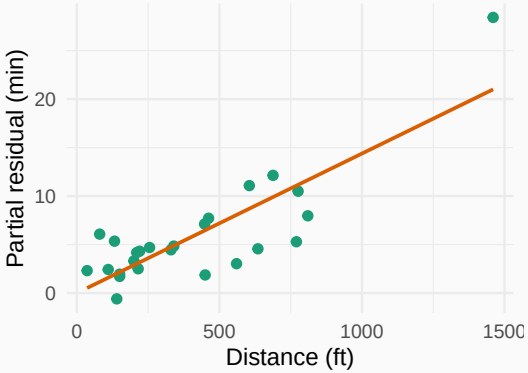
- Linearity: a straight band supports a linear term in $x_j$.
- Points far from the line: outliers after adjusting for the other regressors.
- Points extreme horizontally: potentially influential for $\hat{\beta}_j$.
- A narrow horizontal spread: collinearity between $x_j$ and the other regressors.

# Question: A Point Far from the Line

- A point lies far from the line in the added-variable plot but looks ordinary in the raw scatterplot of $y$ against $x_j$.
- What does this tell you?

# Partial Residual Plot: A Companion

- Plot $e_i + \hat{\beta}_j x_{ij}$ against $x_{ij}$.
- Slope is $\hat{\beta}_j$, as in the added-variable plot.
- The horizontal axis keeps the original scale of $x_j$.
- Curvature in the effect of $x_j$ is easier to see here.



# Limits of Added-Variable Plots

- They show the partial effect of one regressor under a linear model for the others.
- They are not the best tool for deciding which transformation of $x_j$ is needed; use a partial residual plot for that.
- A relationship between $x_j$ and the other regressors that is not linear can distort the picture.
- They are a diagnostic, not a test: the $F$-test of $\beta_j$ remains the formal tool.

# Summary

# Summary of Tests

| Test | Hypothesis | $q$ | Reduced model |
|---|---|---|---|
| Overall | $\beta_1 = \cdots = \beta_k = 0$ | $k$ | Intercept only |
| Individual | $\beta_j = 0$ | 1 | Drop $x_j$ |
| Subset (partial $F$) | $\boldsymbol{\beta}_{(2)} = \mathbf{0}$ | $q$ | Drop $\mathbf{X}_{(2)}$ |
| General linear | $\mathbf{T}\boldsymbol{\beta} = \mathbf{0}$ | $q$ | Impose restrictions |

All use $F_0 = \dfrac{[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})]/q}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n-p)} \sim F_{q,\,n-p}$.

# Key Takeaways

- A full-versus-reduced comparison tests any group of coefficients through one $F$ statistic.
- The extra sum of squares is the drop in $\mathrm{SS}_{\mathrm{Res}}$, with one degree of freedom per added parameter.
- Individual $t$-tests can all be insignificant while a group is highly significant.
- Sequential sums of squares depend on order and add up; partial sums of squares adjust for everything else and do not.
- An added-variable plot shows $\hat{\beta}_j$, the full-model residuals and $\mathrm{SS}_{\mathrm{R}}(\beta_j \mid \text{others})$ in one picture.

# Questions?